Nielsen’s 10 Recommended Heuristics

Evaluating User Experience

- User Experience is a qualitative metric, that is subjected to many factors
- It is an evolving discipline and constantly being updated
- Need to objectively evaluate a design
  - Apple: iOS Human Interface Guidelines (https://developer.apple.com/ios/human-interface-guidelines/)
  - Google: Material Design Guidelines (https://material.io/design)
  - Jakob Nielsen: One of the pioneers who tried to evaluate the user experience on digital platforms with his **heuristic evaluation**

Heuristic Evaluation

- A “discount” usability engineering method. It is quick, cheap, and easy way to evaluate a UI
- Experts examine UI guided by heuristics and record:
  - Problems
  - Severity ratings
  - Solutions (optional)

Nielsen’s 10 Recommended Heuristics

1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

Nielsen’s 10 Recommended Heuristics – In Brief (1 of 2)

1.Visibility of system status
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2.Match between system and the real world
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5.Error prevention
Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

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Nielsen’s 10 Recommended Heuristics – In Brief (2 of 2)

6.Recognition rather than recall
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7.Flexibility and efficiency of use
Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8.Aesthetic and minimalist design
Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9.Help users recognize, diagnose, and recover from errors
Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10.Help and documentation
Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

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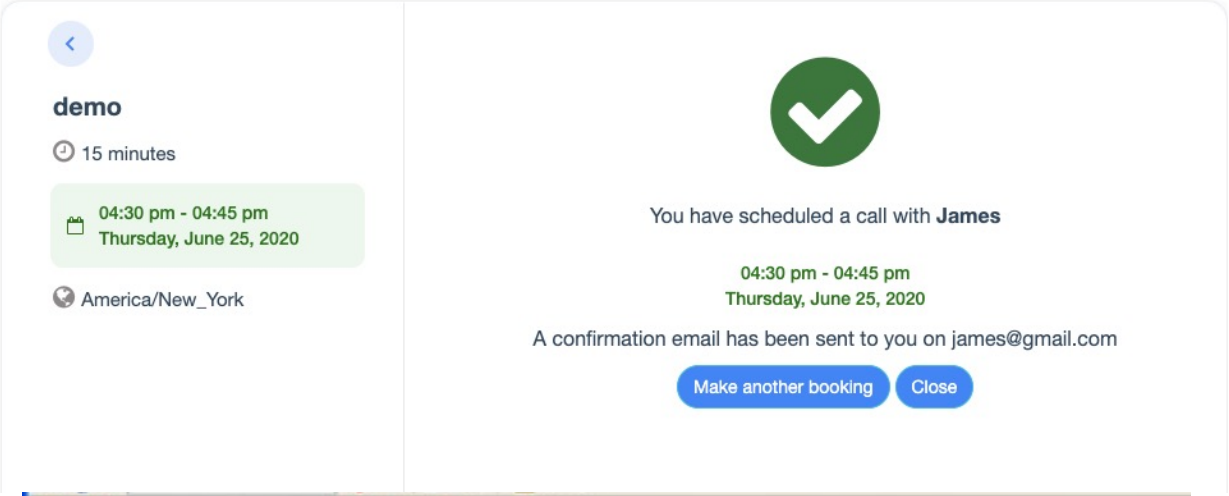
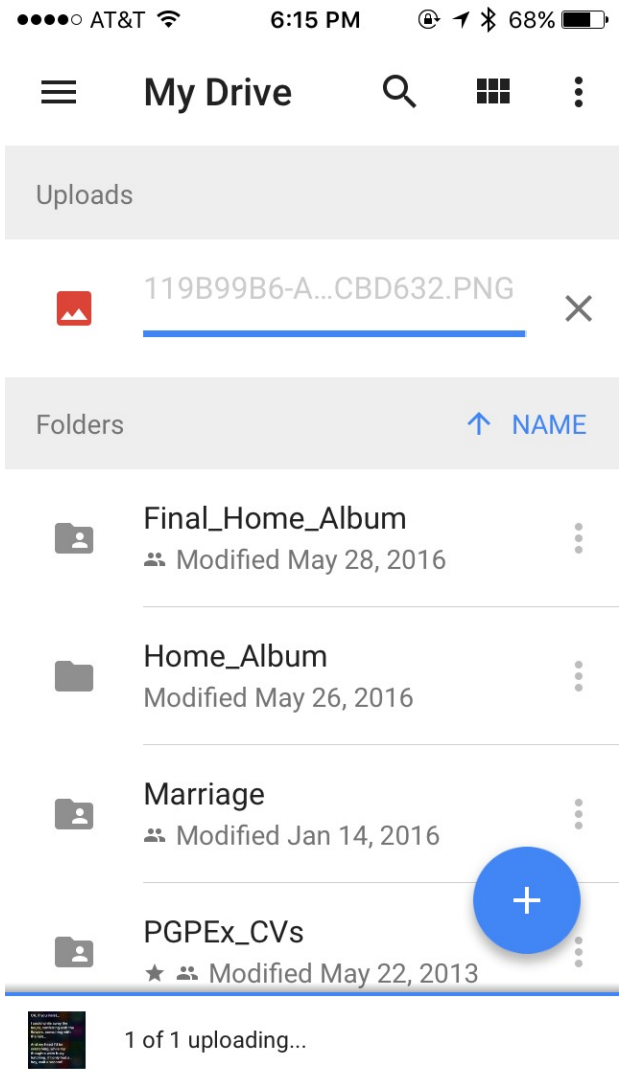
Heuristic 1: Visibility of system status





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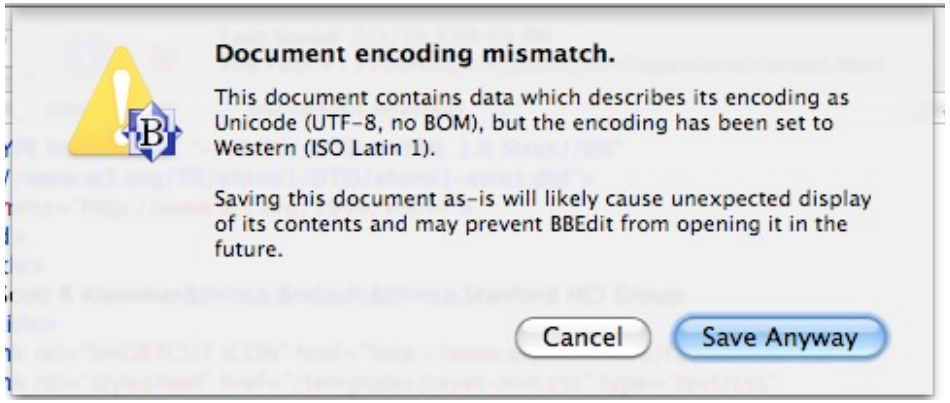
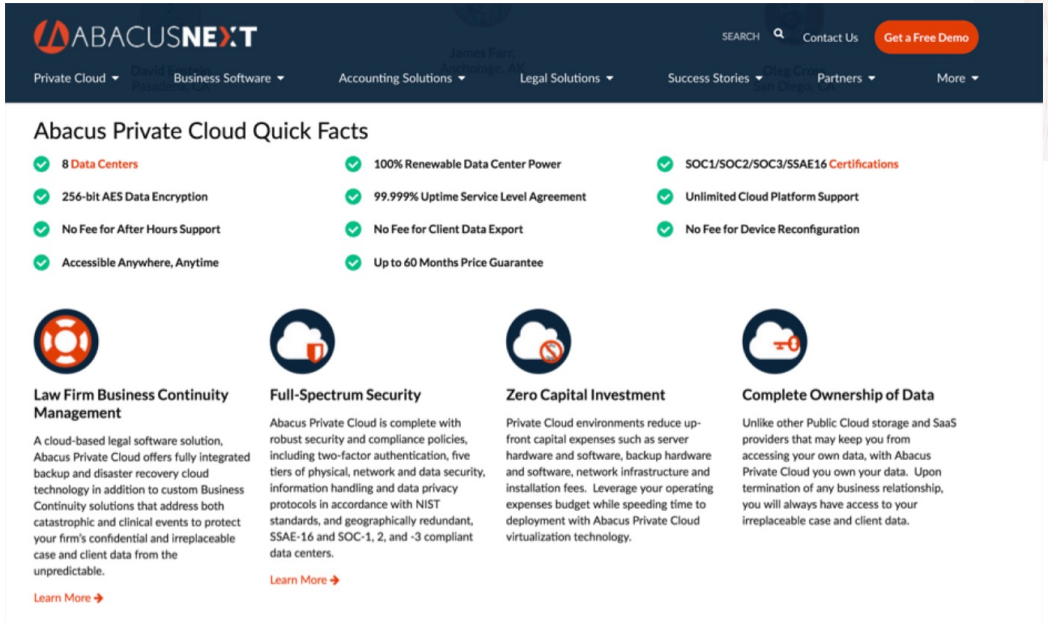
Heuristic 2: Match between system and the real world





← Bad example

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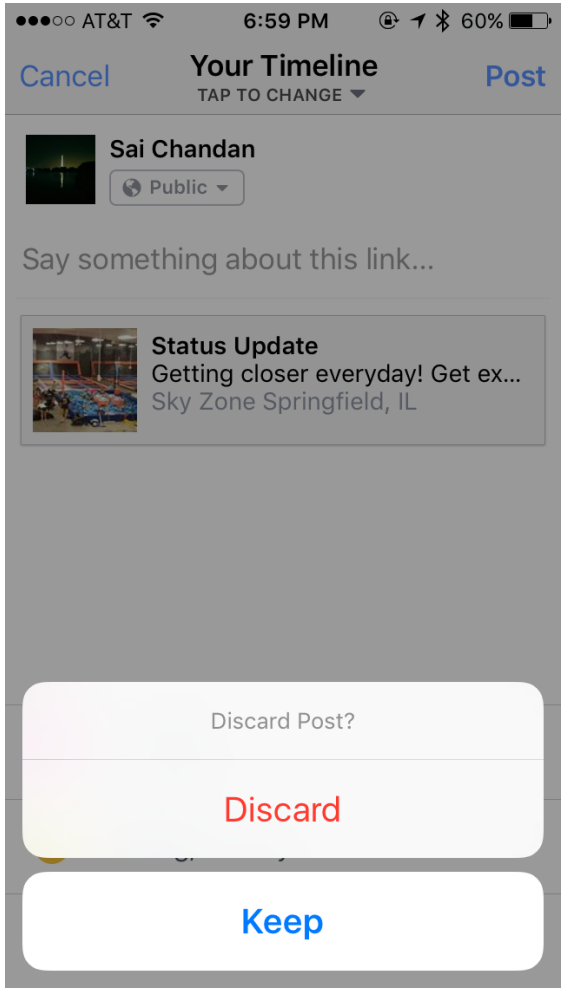
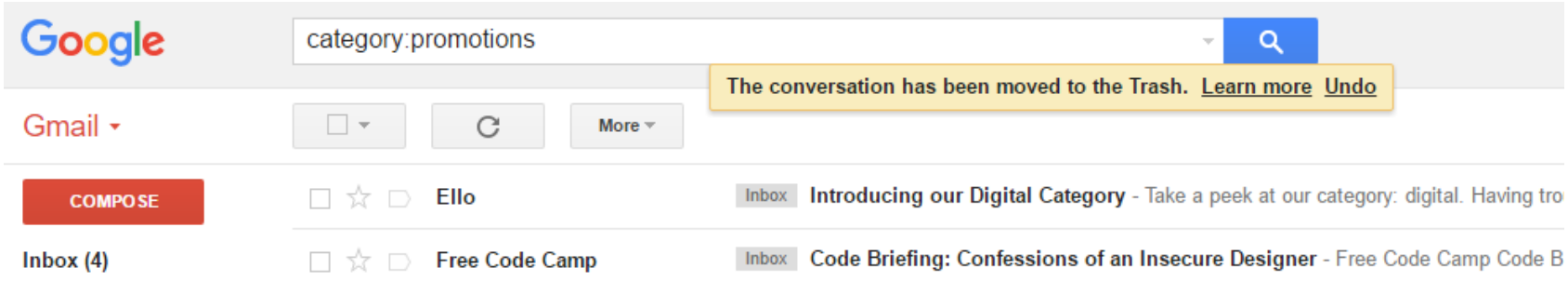
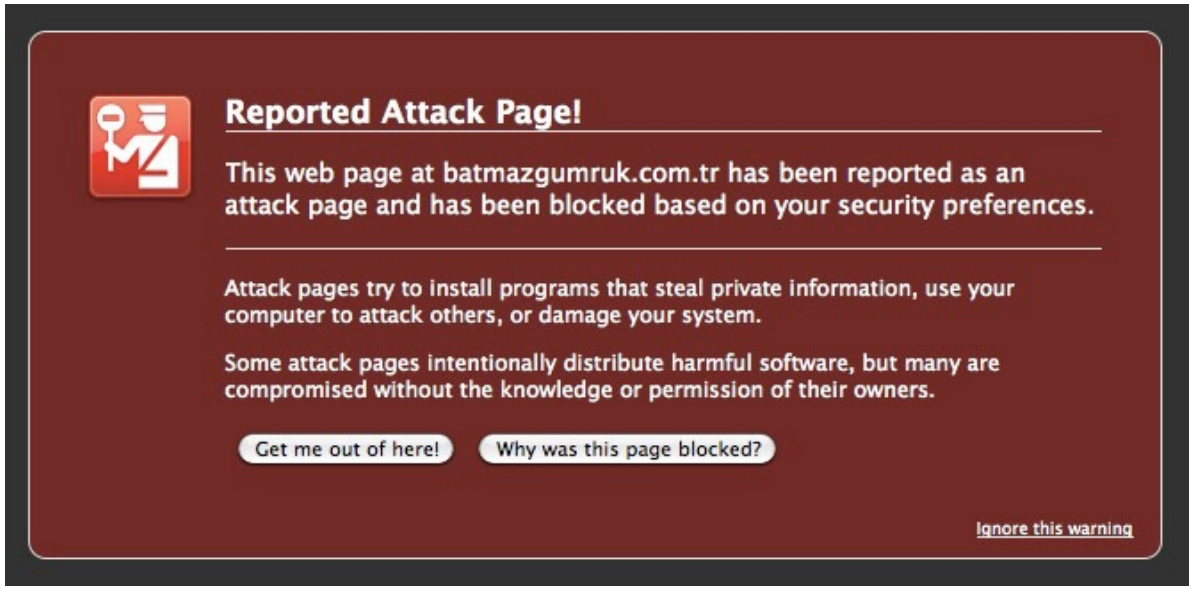
Heuristic 3: User control and freedom







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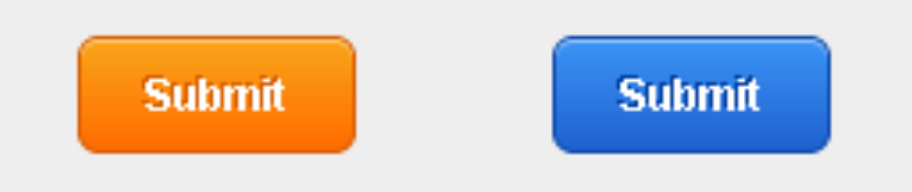
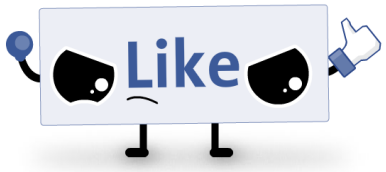
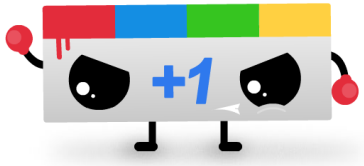
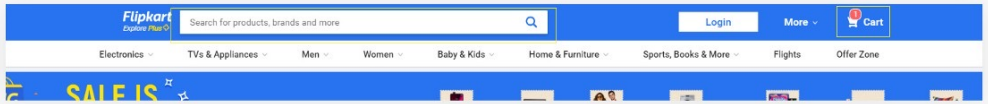
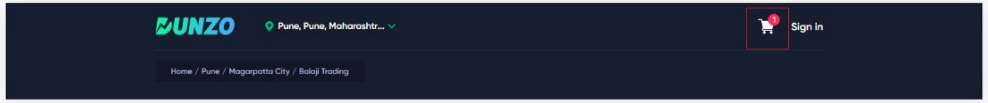
Heuristic 4: Consistency and standards

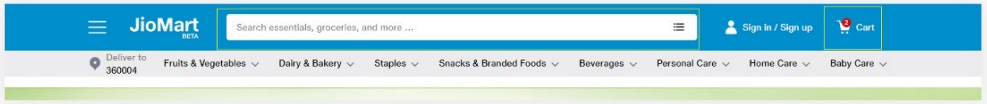


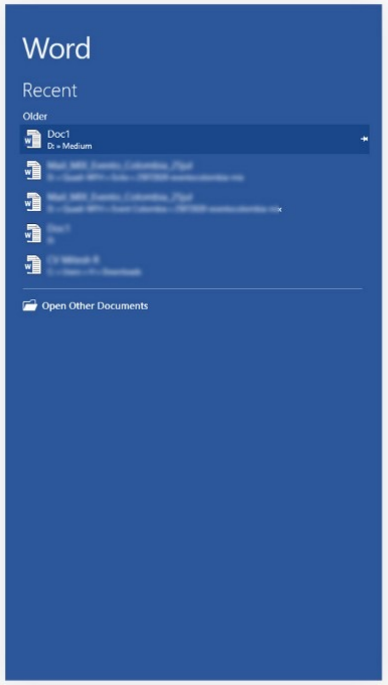


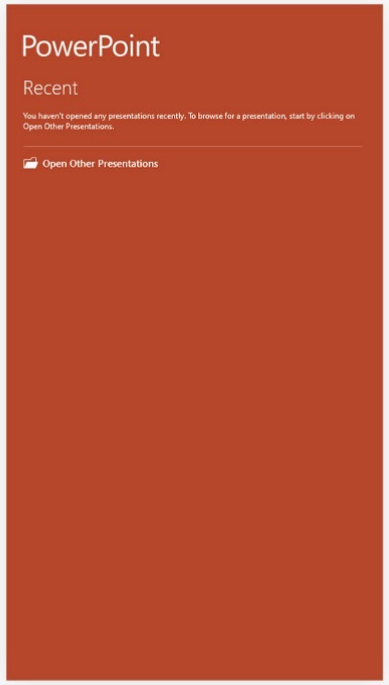












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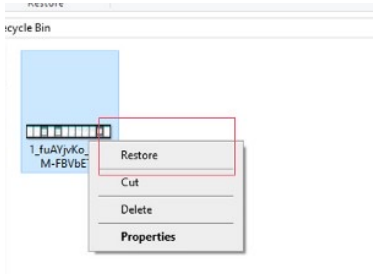
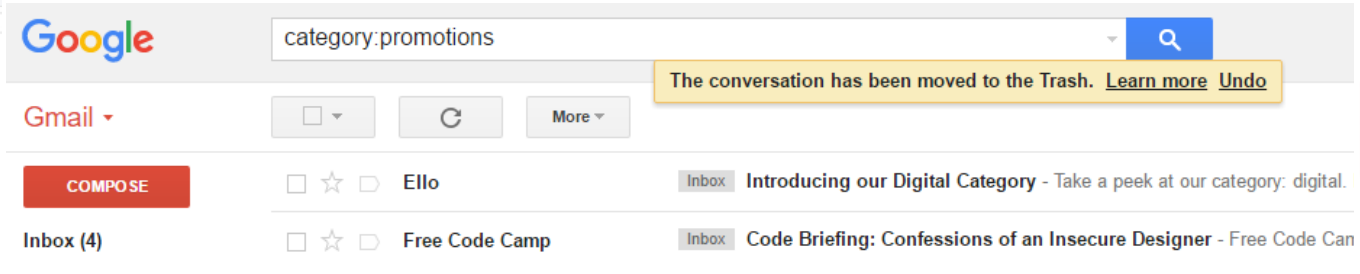
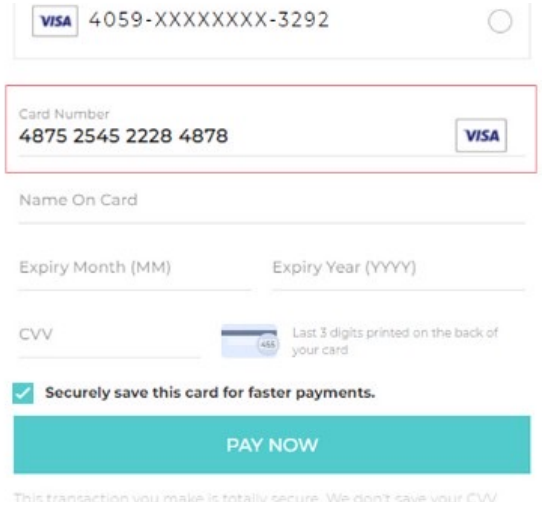
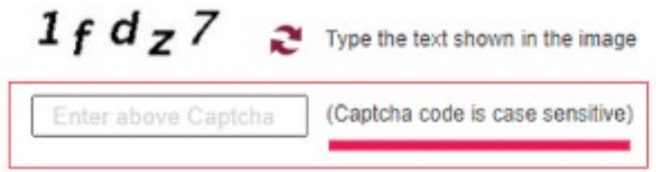
Heuristic 5: Error prevention

Slips

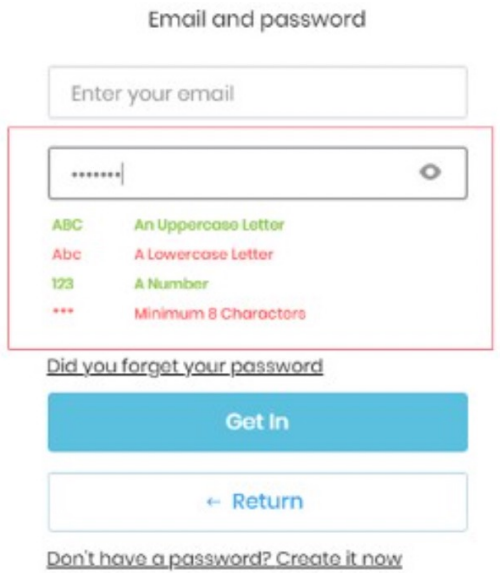


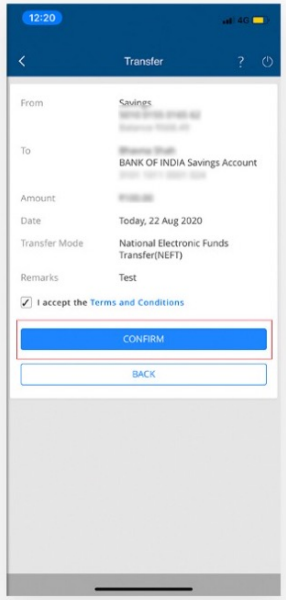


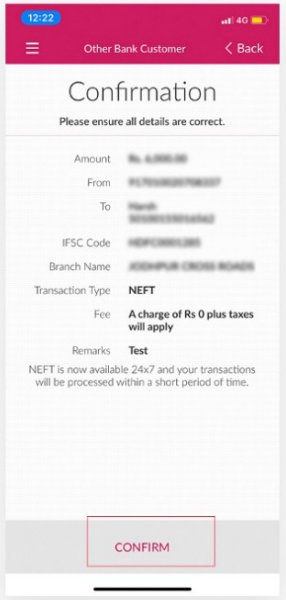


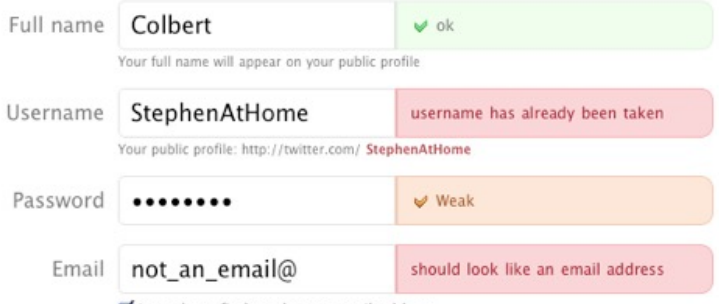


Mistakes









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Nielsen’s 10 Recommended Heuristics – In Brief (2 of 2)

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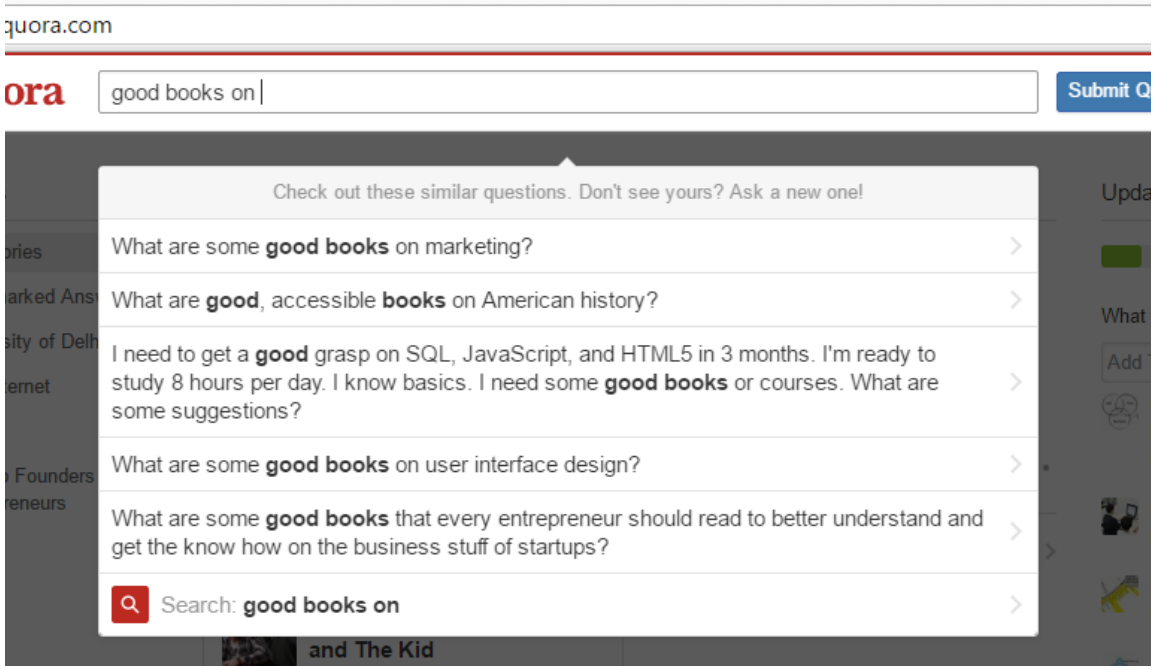
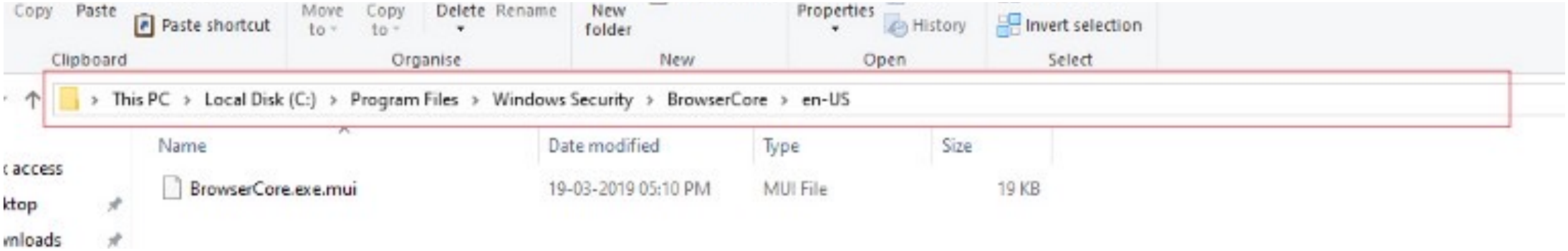
Heuristic 6: Recognition rather than recall





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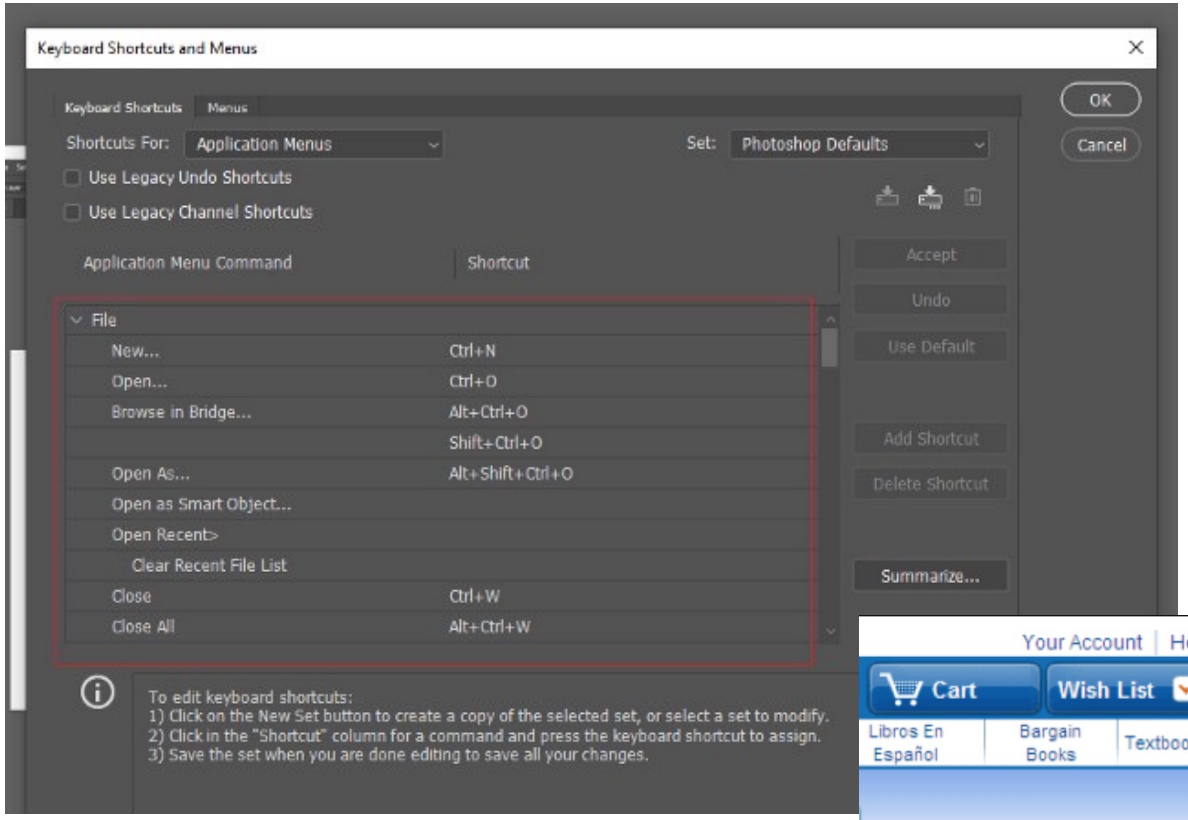

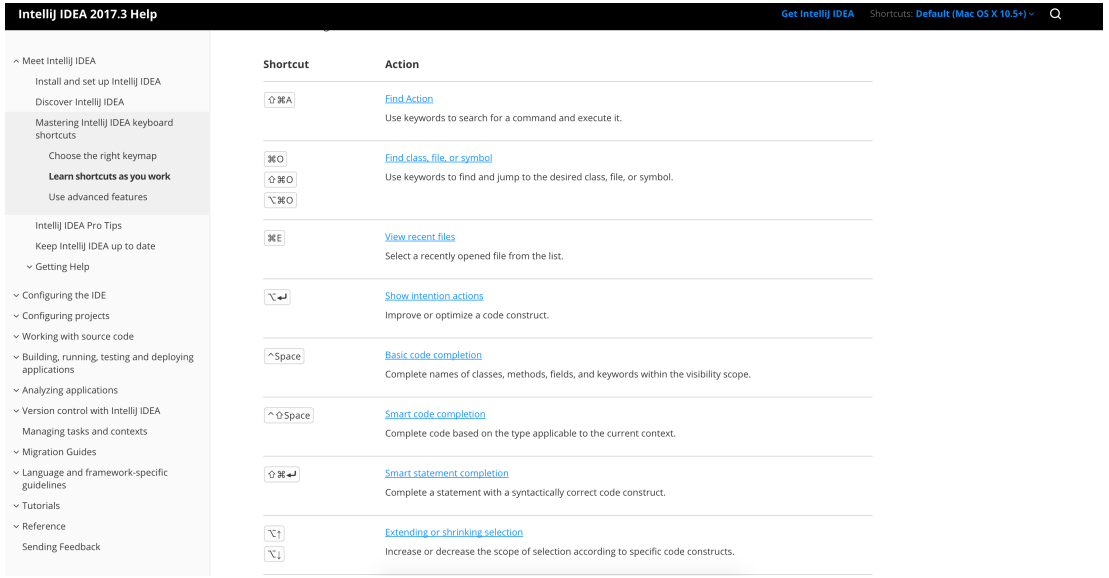

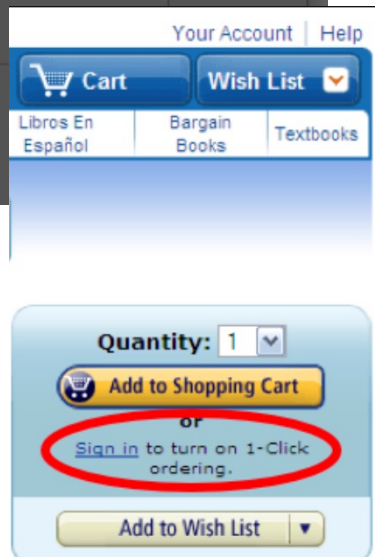

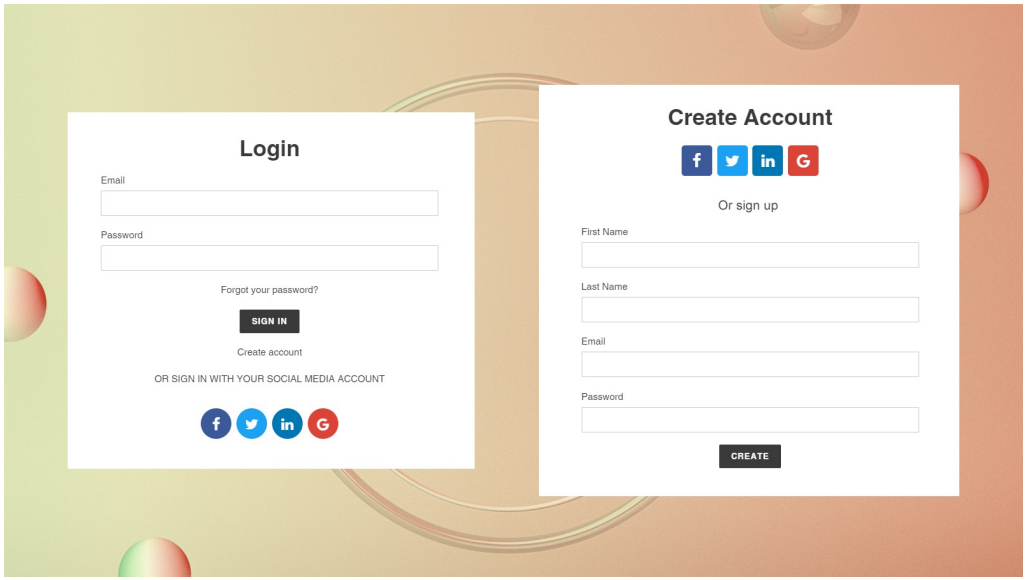

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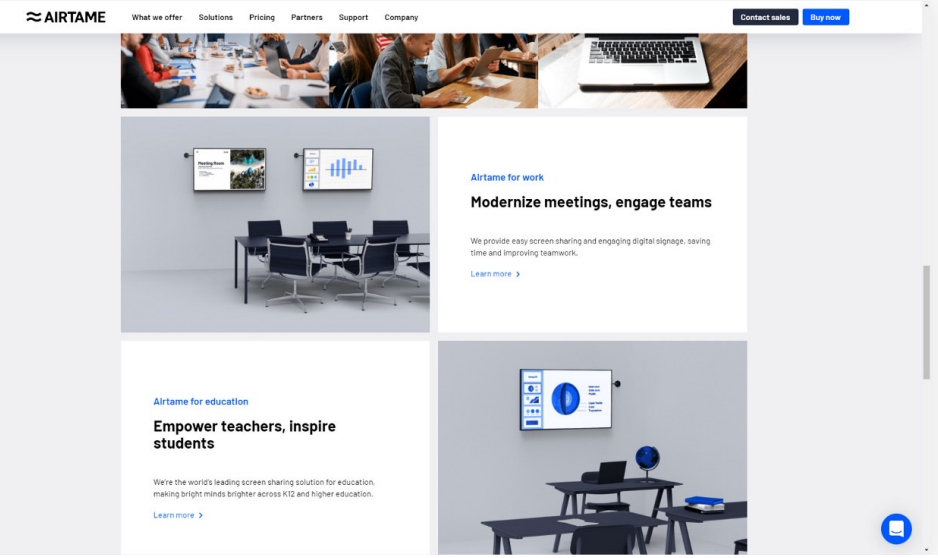
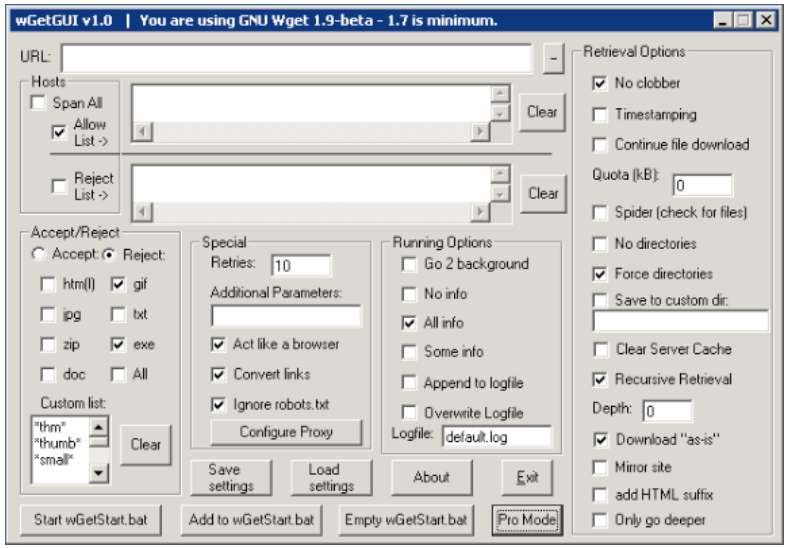
Heuristic 8: Aesthetic and minimalist design





← Bad example

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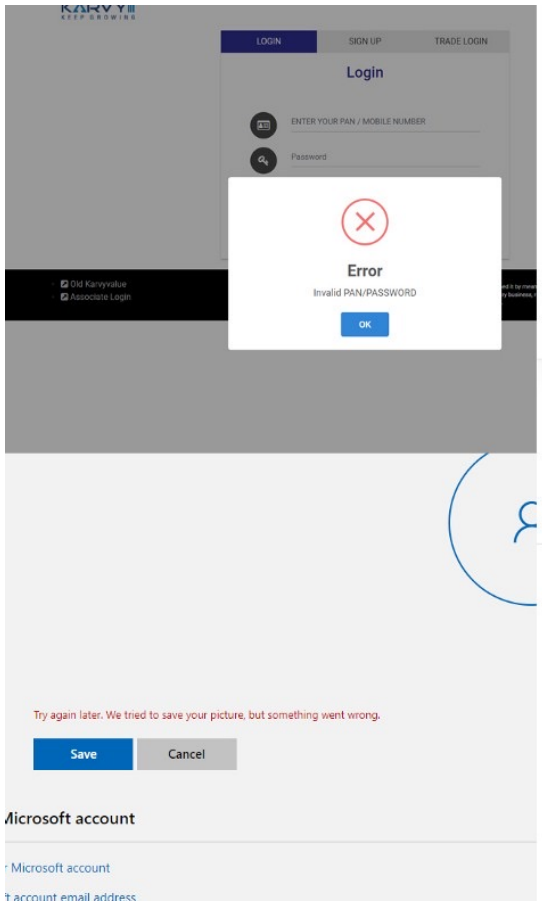
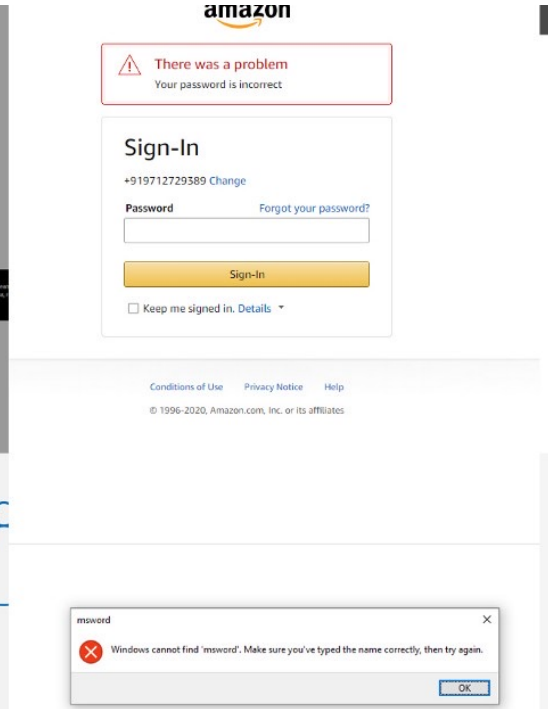
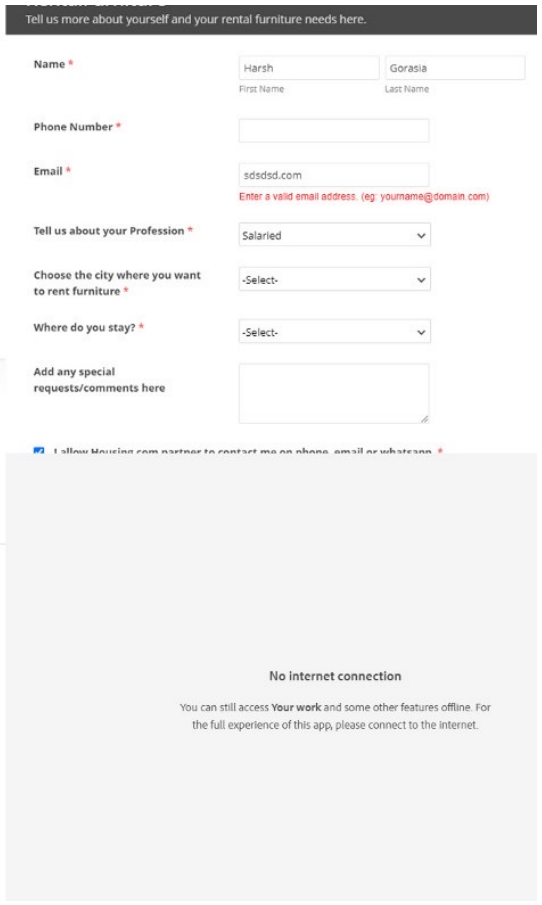
Heuristic 9: Help users recognize, diagnose, and recover from errors







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Nielsen’s 10 Recommended Heuristics – In Brief (2 of 2)

6.Recognition rather than recall
Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

7.Flexibility and efficiency of use
Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8.Aesthetic and minimalist design
Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9.Help users recognize, diagnose, and recover from errors
Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10.Help and documentation
Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

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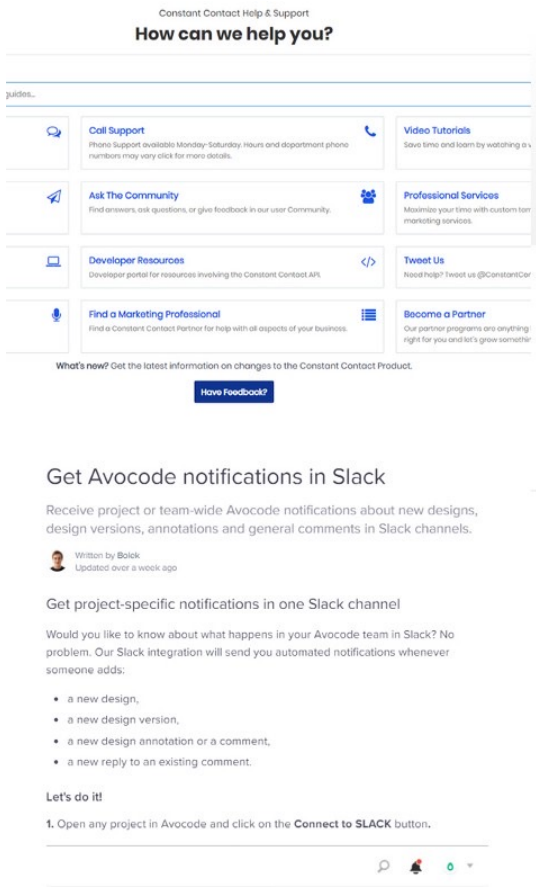
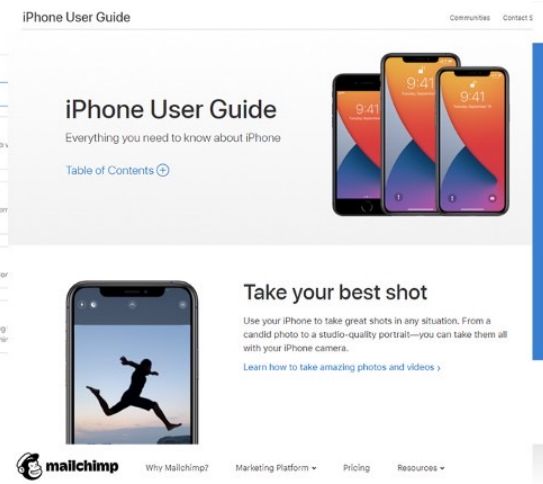
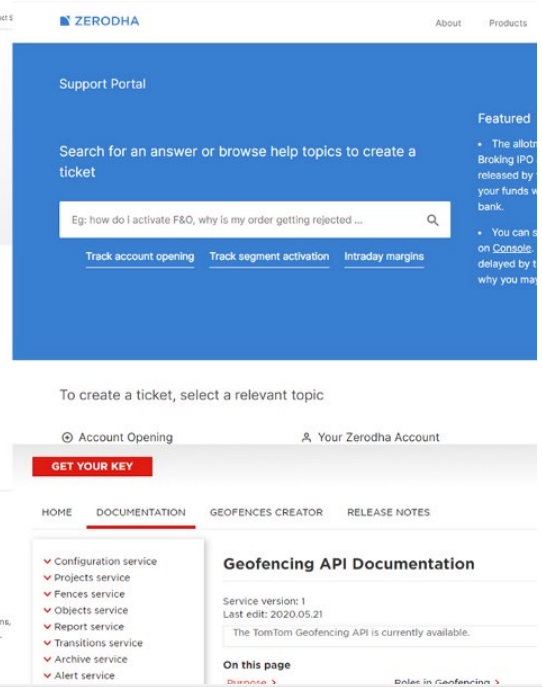
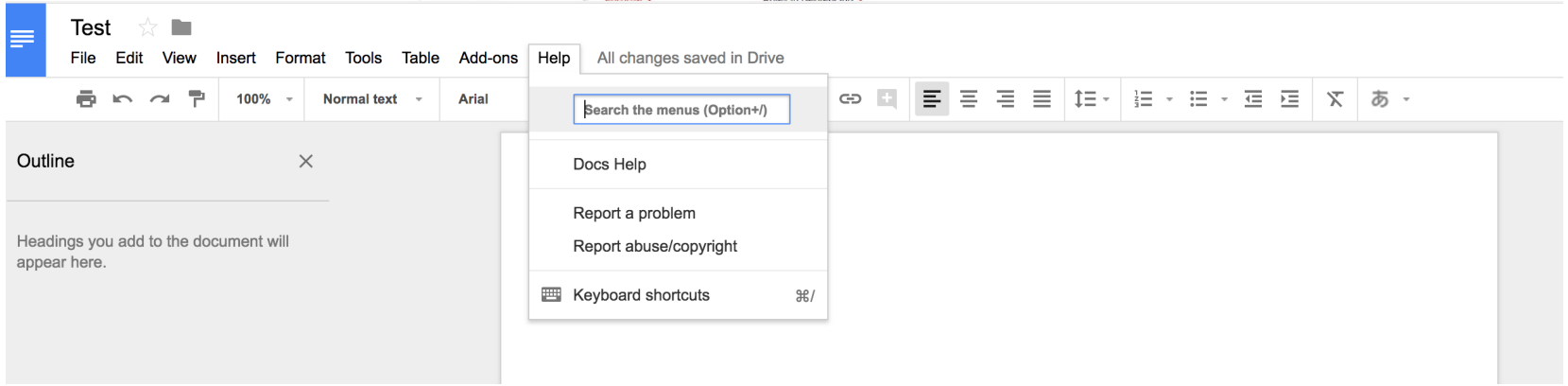
Heuristic 10: Help and documentation









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Thank you

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